

A PRIORI SYMMETRY OF POSITIVE C^2 SOLUTIONS OF THE NONLINEAR STATIONARY SCHRÖDINGER EQUATION IN RADIALLY SYMMETRIC DOMAINS

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ABSTRACT. We study the symmetry of positive solutions to various instances of $-\Delta u + \gamma u \equiv \tau u^3$ in radially symmetric domains of $\mathbb{R}^n$. We will first consider the case of the ball and will prove that any such solution is radially symmetric about the origin of the ball. Ultimately, we show that this extends to Euclidean space, so long as the solutions vanish at infinity.

Throughout this entire text we will consider, in particular, the *Nonlinear Stationary Schrödinger Equation*. Given a particular open domain $\Omega \subseteq \mathbb{R}^n$, $n \geq 2$, we investigate the radial behaviour of positive $u \in C^2(\Omega)$, satisfying

$$-\Delta u + \gamma u \equiv \tau u^{2k+1}, \quad \gamma, \tau > 0, k \in \mathbb{N}_0 \quad (1)$$

provided such a function exists. Especially, this covers the *Nonlinear Stationary Schrödinger Equation*

$$-\Delta u + u \equiv u^3$$

1. THE CASE OF THE COMPACT BALL

In this section we will consider the case where $0 < \rho < \infty$ is given and $\Omega := B_\rho(x_0)$ for some $x_0 \in \mathbb{R}^n$. We will of course assume, without loss of generality, that $x_0 = 0$. We will ultimately show that positive C^2 solutions of

$$\begin{cases} -\Delta u + \gamma u \equiv \tau u^{2k+1} & \text{in } B_\rho(0) \\ u \equiv 0 & \text{on } \partial B_\rho(0) \end{cases} \quad (2)$$

are radially symmetric the *center* of any such ball.

Theorem 1. *Assume u satisfies (2) and the given conditions. Then u is radially symmetric about the origin. Moreover, it is monotonic decreasing about this point.*

PROOF. We will first introduce some notation. Given $\lambda \in (-\rho, \rho)$, we denote by $\mathcal{T}_\lambda$ the plane $x_d \equiv \lambda$ for some component d , fixed. Define Σ_λ the portion of $B_\rho(0)$ to the left of the plane $\mathcal{T}_\lambda$. For any x in the half space $x_d < \lambda$, we will represent the reflection of x across the plane $\mathcal{T}_\lambda$ by x^λ . In other terms,

$$x^\lambda := (x_1, \dots, 2\lambda - x_d, \dots, x_n)$$

Given such a point, we may define the function u evaluated *at the reflection* by

$$u_\lambda(x) := u(x^\lambda)$$

and $w_\lambda := u_\lambda - u$ everywhere. Equipped with these notions, we will divide the proof into 3 steps. First, writing $f(u) := \tau u^{2k+1} - \gamma u$, note that given any $\lambda \in V_\rho(0)$:

$$\Delta w_\lambda - (f(u_\lambda) - f(u)) \equiv 0$$

Fix such λ and consider arbitrary $x \in B_\rho(0)$. We may assume without harm that $x < \rho$. If $u_\lambda(x) = u(x)$, then setting $C(x, \lambda) := 0$ we see that

$$\Delta w_\lambda(x) + C(x, \lambda)w_\lambda(x) = 0$$

Otherwise, using that $f(u) \in C^\infty(\mathbb{R})$, and applying the *Mean Value Theorem*, there exists a constant $C(x, \lambda)$ such that

$$C(x, \lambda) = \frac{f(u_\lambda) - f(u)}{u_\lambda - u}$$

and thus,

$$\Delta w_\lambda(x) + C(x, \lambda)w_\lambda(x) = 0$$

Furthermore, using that $\overline{B_\rho(0)}$ is compact in $\mathbb{R}^n$, we have by continuity of $f'(u)$ over $u(\overline{B_\rho(0)})$ that this function $C(x, \lambda)$ is bounded with compact support. We now commence the *moving plane*.

STEP 1. Here we move the plane to the left of the ball. Let $\lambda \gtrsim -\rho$ and consider the region Σ_λ . For λ sufficiently close to $-\rho$, the region Σ_λ is sufficiently narrow. Of course, we may assume without harm that $\lambda < 0$. We will consider, separately, the two portions of $\partial\Sigma_\lambda$. On $\mathcal{T}_\lambda \cap \partial\Sigma_\lambda$ we trivially have $w_\lambda \equiv 0$. On $\partial B_\rho(0) \cap \partial\Sigma_\lambda$, $u \equiv 0$ but $x^\lambda \in B_\rho(0)$, where u is positive. Whence, we see on this curved part of the boundary, $w_\lambda > 0$ and so, $w_\lambda \geq 0$ on $\partial\Sigma_\lambda$.

Using that our constructed function $C(x, \lambda)$ is bounded, for fixed λ small enough, we may make our region Σ_λ sufficiently narrow, and deduce from the *Narrow Region Principle* that $w_\lambda \geq 0$ in $\overline{B_\rho(0)}$.

STEP 2. Define

$$X := \{\lambda \mid w_\lambda \geq 0 \text{ in } \Sigma_\lambda\} \quad (3)$$

Then, $\sup X \in X$. To see this, we argue by contradiction. Otherwise, there exists a sequence of (λ_i) living in X converging to $\sup X \notin X$. Obviously, by hypothesis, there exists $x_0 \in \Sigma_{\sup X}$ such that $w_{\sup X}(x_0) < 0$. Moreover, using our continuity condition on u there exists a neighbourhood, say $\mathcal{U}$, containing x_0 where $w_{\sup X} < 0$. Observe that each $\lambda_i < \sup X$. Since we can make λ_i arbitrarily close to $\sup X$, we can make it such that for some $x' \in \mathcal{U}$, and λ_i , we have $w_{\lambda_i}(x') < 0$, which is a contradiction.

STEP 3. Here we show $\sup X = 0$. If not, then obviously $\sup X < 0$, we will show we can move the plane forward by a small positive δ , which will contradict the very definition of $\sup X$. Denote, for sake of simplicity, $\Lambda := \sup X$. If $\Lambda < 0$, using Step 2, we also deduce $w_\Lambda \geq 0$ in Σ_Λ . Now, on $\partial\Sigma_\Lambda \cap \partial B_\rho(0)$, the map w_Λ is positive, since the image under the reflection about $\mathcal{T}_\Lambda$ lies in the interior of the open ball. Using the *Strong Maximum Principle*, we must have $w_\Lambda > 0$ in Σ_Λ , since otherwise w_Λ vanishes, and must also vanish on the boundary by continuity. With this in mind, let ℓ denote the maximum width of a so-called *Narrow Region*. Set δ positive such that

$$\delta < \min \left\{ -\Lambda, \frac{\ell}{2} \right\} \quad (4)$$

$$\Omega := \Sigma_{\Lambda+\delta} \cap \left\{ x_d > \Lambda - \frac{\ell}{2} \right\} \quad (5)$$

We now carefully observe that by definition, Ω has with less than ℓ and is thus a *Narrow Region*. We consider all portions of $\partial\Omega$ separately. On the curved portions of this boundary,

using that $\Lambda + \delta < 0$, we have that the reflection about $T_{\Lambda+\delta}$ lies within the interior, yet again. Therefore $w_{\Lambda+\delta} > 0$ here.

Obviously $w_{\Lambda+\delta}$ vanishes on $\mathcal{T}_{\Lambda+\delta}$. We only have to consider $T_{\Lambda-\delta} \subset \partial\Omega$. Notice that all of the previous will hold for any $\delta > 0$ with the upperbounds in (4). Indeed, the plane $\mathcal{T}_{\Lambda-\delta}$ lies within the interior of Σ_Λ , where w_Λ is positive, as was previously shown. By continuity, there exists $\epsilon > 0$ such that $w_{\Lambda+\epsilon} \geq 0$ in this region and especially, over $T_{\Lambda-\delta}$. Letting now $\delta < \epsilon$ as well, we have all of the above and $w_{\Lambda+\delta}$ non negative on this remainder of $\partial\Omega$. Ergo, $w_{\Lambda+\delta} \geq 0$ on $\partial\Omega$. The narrow region principle will imply $w_{\Lambda+\delta} \geq 0$ in this entire region.

Moreover, we also have $w_{\Lambda+\delta} \geq 0$ in Σ_Λ . Therefore, it is clear that $w_{\Lambda+\delta} \geq 0$ in $\Sigma_{\Lambda+\delta}$. Which contradicts our choice of Λ . Therefore, we conclude $\Lambda = 0$.

In light of this same argument and the *Strong Maximum Principle*, we must have $w_\Lambda \equiv 0$ in Σ_Λ , since otherwise we have $w_\Lambda > 0$ in this region, where the same argument will show we can move the plane T_Λ forward again. Thus, we conclude that $u_\Lambda \equiv u$ and that u is symmetric about the origin in the direction of the x_d axis. We can repeat these steps for any finite $n \in \mathbb{N}$, and take their intersection, i.e the origin. It then follows that u is radially symmetric about this point. Monotonicity follows from single variable theory.

The theorem is then proven. ■

2. IN EUCLIDEAN SPACE

Here we will study the same problem, where $\Omega = \mathbb{R}^n$ for $n \geq 2$, as the case $n = 1$ is readily studied in ODE theory. Our boundary condition will be replaced with a decay condition. Essentially, we will again consider positive, C^2 solutions of

$$-\Delta u + \gamma u \equiv \tau u^{2k+1}, \quad \gamma, \tau > 0, k \in \mathbb{N}_0 \quad (6)$$

subject to the additional constraint,

$$\lim_{\|x\| \rightarrow \infty} u(x) = 0 \quad (7)$$

We will ultimately show that the same results apply in this case, where the "origin" is some $x_0 \in \mathbb{R}^n$.

Theorem 2. *Assume $u \in C^2(\mathbb{R}^n)$ is a positive solution (6)-(7). There exists a point $x_0 \in \mathbb{R}^n$ about which u is radially symmetric. That is, for any $x, y \in \mathbb{R}^n$, with $\|x - x_0\| = \|y - x_0\|$, it is implied that*

$$u(x) = u(y)$$

Moreover, u is monotone decreasing about this point.

REMARK. It is useful to observe that the previous argument cannot be applied. Especially since the *Lipschitz* condition does not occur necessarily over all of $\mathbb{R}^n$, and as a result we must resort to other methods.

PROOF. Let u be as given; clearly u is non-trivial. Now, we use that $u, \|\nabla u\| \in \mathcal{O}(e^{-\|x\|})$ and that $g(u) =: f(x) \in \mathcal{O}(e^{-\alpha\|x\|})$ for some given $\alpha > 1$. It is readily seen, by comparison with the function

$$\phi(x) := G(r) \left(1 + \frac{1}{r^s}\right), \quad s > 0 \text{ small} \quad (8)$$

that there exists a constant $\mu_0 > 0$ satisfying for $\|x\| \gg 0$,

$$\|x\|^{\frac{n-1}{2}} e^{\|x\|} u(x) \geq \mu_0 > 0$$

In the above, $r = \|x\|$ and $G(r)$ is an instance of *Green's Function*. Assume now that $\|x\| \rightarrow \infty$ and for some $\xi \in \mathbb{R}^n$

$$\frac{x}{\|x\|} \rightarrow \xi$$

Passing to the limit in the above inequality, we conclude

$$\sqrt{\frac{\pi}{2}} \int_{\mathbb{R}^n} f(y) e^{\xi \cdot y} dy \geq \mu_0 > 0 \quad (9)$$

It can be insightful to note that this limit is particularly easy to see in the case $n = 3$, where the *Green's Function* is given by:

$$G(x) = \frac{e^{-a\|x\|}}{4\pi\|x\|} \quad (10)$$

STEP 1. There exists $\Lambda > 0$ such that $x \in \mathbb{R}^n$ with $x_1 \geq \Lambda$ implies $\partial_{x_1} u(x) < 0$. In order to show this, we assume no such $\Lambda > 0$ exists. In this case, we may construct a sequence (x^i) with $(x_1^i) \rightarrow \infty$ where for each index i , $\partial_{x_1} u(x^i) \geq 0$. Of course, the quotient

$$\frac{x^i}{\|x^i\|}$$

is bounded, and by compactness we may extract a convergent subsequence

$$\frac{x^j}{\|x^j\|} \rightarrow \xi \in \mathbb{R}^n$$

Considering the following limit, we derive a contradiction to equation (9):

$$\frac{\|x^j\|^{\frac{n+1}{2}} e^{\|x^j\|} \partial_{x_1} u(x^j)}{x_1^j} \rightarrow -\sqrt{\frac{\pi}{2}} \int_{\mathbb{R}^n} f(y) e^{\xi \cdot y} dy \geq 0$$

Observing that the relation in (9) holds true for any ξ , we have a contradiction. Therefore, such $\Lambda > 0$ indeed exists. Especially, this shows that any solution u of (6)-(7) is eventually *always strictly decreasing*. Noting that this holds in any direction will shed some light on the wave-like structure of such solutions; especially solutions of the *Stationary Schrödinger Equation*.

STEP 2. There exists $\lambda_0 \geq \Lambda$ such that any $\lambda \geq \lambda_0$ satisfies, for any $x_1 < \lambda$,

$$u(x) > u_\lambda(x)$$

where we adopt the notation of the preceding theorem. Otherwise, there exists a sequence of positive real numbers $(\lambda^i) \rightarrow \infty$ and a corresponding vector sequence (x^i) with $x_1^i < \lambda^i$ and $u(x^i) \leq u_{\lambda^i}(x^i)$. Let us denote by z^i the reflection of x^i about the static plane $x_1 \equiv \lambda^i$. That is,

$$z^i := (2\lambda^i - x_1^i, \dots, x_n)$$

Observe that $x_1^i < \Lambda$ for all indices, since otherwise we apply Step 1 to notice

$$u(z^i) - u(x^i) = \int_{x^i}^{z^i} \partial_{x_1} u(\sigma) d\sigma_1 < 0$$

Again, using that boundedness of the quotient sequence $\frac{x^i}{\|x^i\|}$ for some $\xi \in \mathbb{R}^n$:

$$\frac{x^j}{\|x^j\|} \rightarrow \xi$$

Moreover, we take note that $\xi_1 \leq 0$ since

$$\frac{x^j}{\|x^j\|} < \frac{\Lambda}{\|x^j\|} \rightarrow 0$$

- (1) If $\xi_1 = 0$, with the help of Step 1, for $\lambda^i \geq \lambda$ we must have

$$u(x^i) \leq u(z^i) \leq u(x_\lambda^i)$$

In which case, taking the limit of:

$$0 \geq \lim_{\lambda \rightarrow x_1^j} \frac{\|x^j\|^{\frac{n+1}{2}} e^{\|x^j\|}}{\lambda - x_1^j} \left(u(x_\lambda^j) - u(x^j) \right) = \sqrt{2\pi} \int_{\mathbb{R}^n} e^{\xi' \cdot y'} (\lambda - y_1) dy$$

However, since $f(y) \in \mathcal{O}(e^{-\alpha\|y\|})$ for $\alpha > 1$, we can make $\lambda > 0$ very large, hence contradicting (9) yet again. We then deduce $\xi_1 = 0$ is impossible

- (2) If $\xi_1 < 0$, we set $d := \|z^j\| - \|x^j\|$. Then, by the very definition of z^i , we see that $\|z^i\| \rightarrow \infty$ and thus $u(z^i) \rightarrow 0$ by our condition in (7). Using that $0 < u(x^i) \leq u(z^i)$ we deduce from the *Squeeze Theorem* that $u(x^i) \rightarrow 0$ as well. Furthermore, since $u > 0$ and of class C^2 in $\mathbb{R}^n$, we must have then that $\|x^i\| \rightarrow \infty$.

A direct calculation can verify that

$$d^2 + 2d\|x_1^j\| = 4(\lambda^j)^2 - 4\lambda^j x_1$$

Since we can make $x_1^j \approx \|x^j\|\xi_1$, we note that $d \rightarrow \infty$ in j . We may assert that for sufficiently large j :

$$\begin{aligned} 0 < \|x^j\|^{\frac{n-1}{2}} e^{\|x^j\|} u(x^j) &= \left(\frac{x^j}{z^j} \right)^{\frac{n-1}{2}} e^{\|x^j\| - \|z^j\|} \|z^j\|^{\frac{n-1}{2}} e^{\|z^j\|} u(x^j) \\ &\quad \left(\frac{x^j}{z^j} \right)^{\frac{n-1}{2}} e^{\|x^j\| - \|z^j\|} \|z^j\|^{\frac{n-1}{2}} e^{\|z^j\|} u(x^j) \end{aligned}$$

Since this latter term approaches 0, we see the former does as well. This contradicts (9) and hence we obtain a global contradiction.

STEP 3. Let $X := \{\lambda \mid x_1 < \lambda \implies u(x) > u_\lambda(x)\}$. X is open. We argue by contradiction. If this is not the case, then for some $\lambda \in X$, given any $\epsilon > 0$, we may assert that

$$B(\lambda, \epsilon) \not\subseteq X$$

This is true in particular for $\epsilon := \frac{1}{n} > 0$, for $n \in \mathbb{N}$. For any such n , using $B(\lambda, 1/n) \not\subseteq X$, there exists a point λ^n , with $|\lambda - \lambda^n| < \frac{1}{n}$ where for some point x , say, x^n with $x_1^n < \lambda^n$ we have $u(x^n) \leq u(x^{n^{\lambda^n}})$. In this way we construct sequences (λ^i) , (x^i) satisfying $(\lambda^i) \rightarrow \lambda$ and $x_1^i < \lambda^i$ but $u(x^i) \leq u(z^i)$, for $z := x^{i^{\lambda^i}}$.

Furthermore, $\|x^i\| \rightarrow \infty$. For suitable subsequences and $\xi, \zeta \in \mathbb{R}^n$, with $\xi_1 \leq 0$ as before,

$$\frac{x^j}{\|x^j\|} \rightarrow \xi, \quad \frac{z^j}{\|z^j\|} \rightarrow \zeta$$

Suppose that $\xi_1 \leq 0$, it is easy to verify by calculation that

$$\|x_\lambda\| - \|x\| \equiv \frac{4\lambda(\lambda - x_1)}{\|x_\lambda\| + \|x\|} \quad (11)$$

Sending $j \rightarrow \infty$ we see especially that

$$\|x^j\| - \|z^j\| \rightarrow 2\lambda\xi_1$$

In which case, using again that

$$\|x^j\|^{\frac{n-1}{2}} e^{\|x^j\|} u(x^j) \leq \left(\frac{\|x^j\|}{\|z^j\|} \right)^{\frac{n-1}{2}} e^{\|x^j\| - \|z^j\|} \|z^j\|^{\frac{n-1}{2}} e^{\|z^j\|} u(z^j)$$

it becomes obvious that

$$\int_{\mathbb{R}^n} f(y) e^{\xi \cdot y} dy \leq e^{2\lambda\xi_1} \int_{\mathbb{R}^n} f(y) e^{\xi \cdot y} dy \quad (12)$$

We will here consider the half-space $\mathbb{S} := \mathbb{R}^{n-1} \times \{x_1 < \lambda\}$. Set the map $\omega_\lambda := u - u_\lambda > 0$ in $\mathbb{S}$ by hypothesis that $\lambda \in X$. Moreover, we can calculate

$$(\Delta - 1)\omega_\lambda = \tau u_\lambda(x)^3 - \tau u(x)^3 \leq 0 \quad (13)$$

$$\because D\tau(\cdot)^3 \geq 0 \text{ in } \mathbb{R} \quad (14)$$

Using a comparison function of the form

$$\psi := \sqrt{\frac{\pi}{2}} \left(\frac{e^{-r} x_1}{r^{\frac{n+1}{2}}} \right) \left(1 + \mathcal{O}\left(\frac{1}{r}\right) \right)$$

it can be shown that for some $\mu > 0$, given $x \in \mathbb{S}$ it follows

$$\frac{\|x\|^{\frac{n+1}{2}} e^{\|x\|}}{\lambda - x_1} \omega_\lambda \geq \mu > 0 \quad (\star)$$

If $\xi_1 = 0$, consider the lefthand of $(\star)$ yields

$$\int_{\mathbb{R}^n} f(y) e^{\xi \cdot y} (\lambda - y_1) dy \leq 0$$

which contradicts $(\star)$ itself. Otherwise we must have $\xi_1 < 0$. In this case, note that since $\|x^j\| \rightarrow \infty$, $x_1^j \rightarrow -\infty$, and therefore for sufficiently large j the subsequence (x_1^j) lies in the half space $\mathbb{S}$. Consequently,

$$\mu \leq \frac{\|x^j\|^{\frac{n+1}{2}} e^{\|x^j\|} (u(x^j) - u(z^j))}{\lambda - x_1^j} \quad (15)$$

$$= \frac{\|x^j\|}{\lambda - x_1^j} \left(\|x^j\|^{\frac{n-1}{2}} e^{\|x^j\|} u(x^j) - \|x^j\|^{\frac{n-1}{2}} e^{\|x^j\|} u(z^j) \right) \quad (16)$$

$$< \frac{-\|x^j\|}{x_1^j} \left(\|x^j\|^{\frac{n-1}{2}} e^{\|x^j\|} u(x^j) - \left(\frac{\|x^j\|}{\|z^j\|} \right)^{\frac{n-1}{2}} e^{\|x^j\| - \|z^j\|} \|z^j\|^{\frac{n-1}{2}} e^{\|z^j\|} u(z^j) \right) \quad (17)$$

where taking the limit, we see

$$-\frac{1}{\xi_1} \int_{\mathbb{R}^n} (e^{\xi \cdot y} - e^{2\lambda\xi_1 + \xi \cdot y}) dy \geq \mu \quad (18)$$

contradicting (12).

STEP 4. If for λ , $x_1 < \lambda$ implies $u(x) \geq u_\lambda(x)$ with $u \not\equiv u_\lambda$, then $u(x) > u_\lambda(x)$ everywhere. To see this, set

$$L[\cdot] := (-\Delta + 1) \geq 0$$

and apply the *Strong Maximum Principle* to $z := u - u_\lambda \geq 0$.

STEP 5. Here we complete the proof of the theorem. Using the previous steps, in particular the third, we see there exists a maximal interval of λ , $\lambda_* < \lambda \leq \infty$, for which $u > u_\lambda$ when $x_1 < \lambda$. Of course, $\lambda_* \neq -\infty$. Considering the implications of Step 4, we must have $u \equiv u_\lambda$ in $\mathbb{R} \times \{x_1 < \lambda_*\}$, since otherwise we have the stronger result $u > u_\lambda$ in this half-space, whereby a continuity argument shows that we may extend this interval further, which would contradict our choice of maximal interval.

We can repeat this process in any direction, and take the intersection of all λ_d as the origin. It then follows that u is radially symmetric about this point x_0 . The theorem is then proven. ■

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